Assignment-4.4

“Cable types and tools”

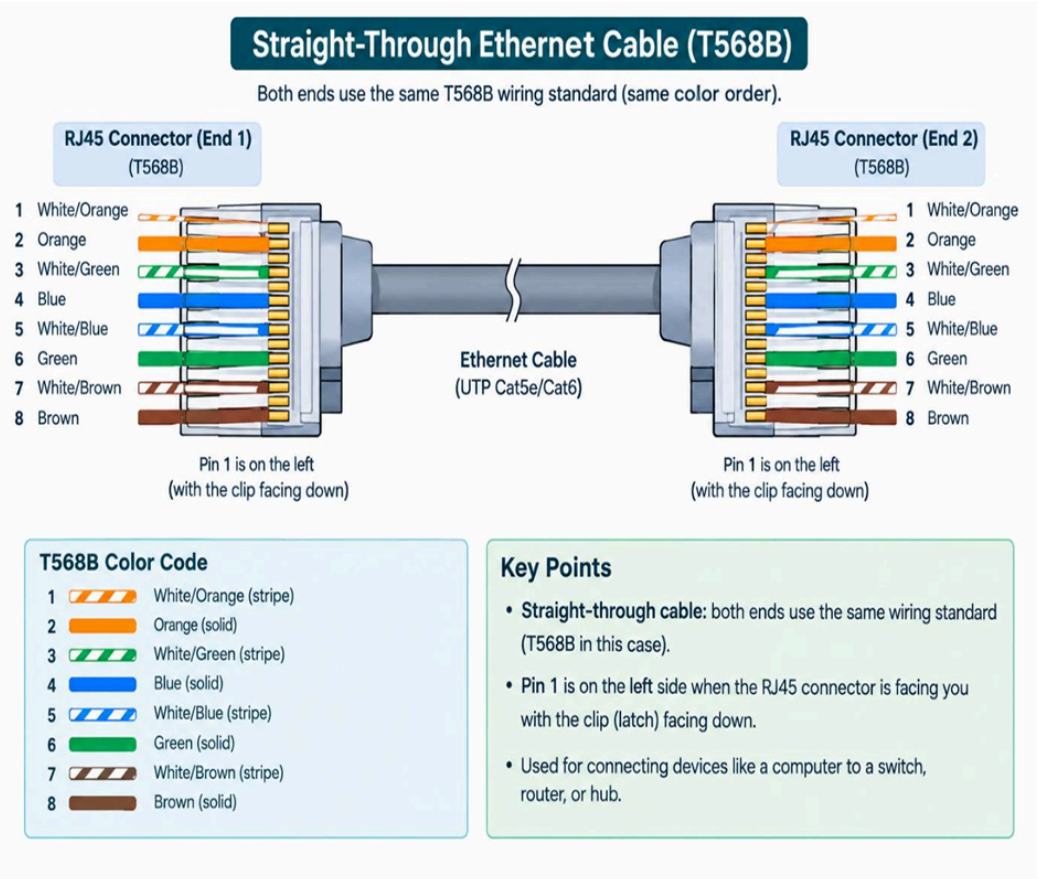
TASK-1 List 3 key differences between Ethernet cables and fiber optic cables, and give one real-world example where each type would be used (for example, home WiFi router, data center, college campus backbone).

Basis of Difference	Ethernet Cable	Fiber-Optic Cable
1. Data Transmission	Uses electrical signals to transmit data through copper wires.	Uses light signals to transmit data through thin glass or plastic fibers.
2. Speed and Distance	Supports speeds up to 10 Gbps over shorter distances, generally limited to 100 meters.	Supports very high speeds and can carry data over much longer distances.
3. Cost and Installation	Affordable and easy to install in homes, offices, and computer labs.	Costs more and requires specialized installation tools and skills.

Ethernet Cable: An Ethernet cable connects a desktop computer to a home Wi-Fi router to provide a stable wired internet connection.

Fiber-Optic Cable: Fiber-optic cables connect different buildings across a college campus or link equipment in a data center to transfer large amounts of data over long distances

2.Draw and label a diagram showing the internal wiring of a straight-through Ethernet cable (use T568B wiring standard), indicating the color of each wire and its pin number.



TASK-3.Watch a YouTube video demonstrating how to use an RJ45 crimper, then write a step-by-step guide (in your own words) on how to attach an RJ45 connector to a CAT6 cable.

An RJ45 connector is the plastic plug commonly used at the end of an Ethernet cable. An RJ45 crimping tool is used to secure the connector to the cable so that it can connect to a router, switch, or computer.

For this task, I would follow a demonstration video and use the T568B wiring standard.

YouTube Video Source: Search for CAT6 RJ45 Crimping Tutorials

Step-by-Step Guide

Step 1: Collect the Required Tools

Before starting, gather the following items:

- CAT6 Ethernet cable
- RJ45 connector suitable for the CAT6 cable
- RJ45 crimping tool
- Wire stripper or cable jacket stripper
- Ethernet cable tester

These tools help prepare the cable, attach the connector, and check whether the wiring is correct.

Step 2: Cut the Cable

Cut the CAT6 cable to the required length using a suitable cable cutter. Make sure the end is clean and even so the wires can be prepared properly.

Step 3: Remove the Outer Jacket

Use a cable stripper to carefully remove a small section of the outer cable jacket. Avoid damaging the insulated wires inside because damaged wires can cause connection problems.

Step 4: Arrange the Wires

Separate the eight insulated wires and arrange them in the T568B order:

1. White-Orange
2. Orange
3. White-Green
4. Blue
5. White-Blue
6. Green
7. White-Brown
8. Brown

Make sure the wires are straight and in the correct order.

Step 5: Trim the Wires

Carefully trim the wires evenly so they can fit into the RJ45 connector. Keep enough length for the wires to reach the connector contacts, following the connector manufacturer's instructions.

Step 6: Insert the Wires into the RJ45 Connector

Insert the wires into the connector while maintaining the T568B order. Check that each wire reaches the appropriate contact position and that the cable jacket extends far enough into the connector to be held securely.

Step 7: Crimp the Connector

Place the RJ45 connector into the correct slot of the crimping tool. Squeeze the handles firmly as instructed by the tool manufacturer to secure the contacts and connector.

Step 8: Test the Cable

Use an Ethernet cable tester to check the completed cable. The tester should show the correct pin connections from pin 1 through pin 8.

If the wiring is incorrect or a connection is missing, inspect the cable and correct the termination before using it.

Conclusion

Attaching an RJ45 connector requires the correct tools and wire arrangement. Following the T568B standard at both ends creates a straight-through Ethernet cable suitable for common network connections.

TASK-4.Imagine you are setting up a gaming café with 10 PCs for LAN gaming. Decide which cable type (Ethernet or fiber optics) and which tool (RJ45 crimper or fiber cleaver) you would use, and explain your choices in 3-4 sentences.Consider speed, distance, and budget for your answer.

For a gaming café with 10 PCs, I would choose **CAT6 Ethernet cables** because they provide a fast, stable connection and are suitable for short distances. I would connect all 10 PCs to a network switch and connect the switch to the router for internet access. I would use an **RJ45 crimping tool** to attach RJ45 connectors to the cable ends when making custom-length cables. Fiber-optic cables and a fiber cleaver would not be necessary because Ethernet cables are more affordable and easier to install for this small gaming setup.